

CALEB LAREAU

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CURRENT APPOINTMENT

Assistant Member , Program in Computational and Systems Biology Memorial Sloan Kettering Cancer Center (New York, NY)	2023 - Present
Assistant Professor , Computational Biology and Medicine Weill Cornell Medical School (New York, NY)	2023 - Present

EDUCATION AND TRAINING

Postdoctoral Fellowship , Departments of Pathology and Genetics Stanford University (Palo Alto, CA)	2023
Doctor of Philosophy , Biological and Biomedical Sciences Harvard University (Cambridge, MA)	2020
Master of Arts , Biostatistics Harvard University (Cambridge, MA)	2017
Bachelor of Science , <i>summa cum laude</i> , Biochemistry and Mathematics University of Tulsa (Tulsa, OK)	2015

AWARDS AND HONORS

Pershing Square Sohn Cancer Research Alliance Prize	2026
National Academy of Medicine - Catalyst Awardee	2025
Michelson Medical Research Foundation: Next Generation Prize	2024
Clarivate Highly Cited Researcher (Cross-Field)	2024, 2025
Society for Immunotherapy of Cancer - Sparkathon Young Investigator	2023
Memorial Sloan Kettering Emerging Leader in Computational Oncology	2023
Gladstone Institutes Distinguished Achievement in Science Award (Genomic Immunology)	2023
National Human Genome Research Institute - Pathway to Independence Award (K99/R00)	2022
STAT Wunderkind	2022
Forbes 30 Under 30 - Science	2022
Parker Institute for Cancer Immunotherapy - Parker Scholar	2021
Broad Institute Next Generation in Biomedicine	2021
10x Genomics Featured Scientist in Innovator Series	2021
Stanford Science Fellowship	2020
Deutsche Forschungsgemeinschaft Mercator Fellowship	2020
Ruth L. Kirschstein National Research Service Award Individual Predoctoral Fellowship (F31)	2018
Harvard University Certificate of Distinction in Teaching	2017
National Science Foundation Graduate Research Fellowship (NSF-GRFP)	2015
Phi Kappa Phi Walter and Adelheid Hohenstein Fellow	2015
Pi Kappa Alpha Robertson Most Outstanding Undergraduate Student	2014
Jess Choteau Outstanding Senior	2014
Deutscher Akademischer Austauschdienst RISE Scholar	2013, 2014
Barry M. Goldwater Scholar	2013
University of Tulsa Presidential Scholar	2011
Oklahoma Medical Research Foundation Sir Alexander Fleming Scholar	2011

TEACHING EXPERIENCE

Gerstner Sloan Kettering Graduate School

Lecturer, Introduction to Artificial Intelligence

Spring 2026- Present

Lecturer, Basic Science Module; Cancer Biology and Cancer Engineering Program

Fall 2024- Present

Weill Cornell Medicine

Course Co-Head, Computational Genomics

Spring 2025- Present

Lecturer, Deep Learning in Biology and Medicine

Spring 2025- Present

Lecturer, Functional Interpretation of High-Throughput Data

Spring 2024- Present

Faculty Moderator, Computational Biology and Medicine Journal Club

Fall 2023- Present

Stanford University

Course Head; Cell, Gene, and Immune Therapies

Spring 2023

Lecturer; Single-cell Immunogenomics

Spring 2022, 2023

Lecturer; Introduction to Applied Computational Tools in Immunology

Winter 2022, 2023

Harvard University

Teaching Associate; Introduction to Computational Biology and Bioinformatics

Spring 2017, 2020

Teaching Associate; Introduction to Statistical Genetics

Fall 2016, 2017

Cold Spring Harbor Laboratory

Lecturer; Advanced Sequencing Technologies

Fall 2019

University of Tulsa

Teaching Associate; Molecular Modeling

Spring 2014, 2015

Teaching Associate; Energy Technology for the Future

Spring 2013

ACADEMIC SERVICE

OPEN SCIENCE

bioRxiv Affiliate

2025- Present

ADMISSIONS

Tri-institutional MD/PhD Admissions Committee Member

2026- Present

Tri-institutional Computational Biology and Medicine Admissions Committee Member

2026- Present

Gerstner Sloan Kettering Cancer Engineering Admissions Committee

2025- Present

OUTREACH

Volunteer, New York Academy of Sciences Career Fair (representing MSKCC)

2025-Present

Mentor/Lecturer, MSKCC Computational Biology Summer Program

2024-Present

Mentor, Weill Cornell High School Catalyst Program

2024-Present

Goldwater Scholar Mentorship Program

2016-Present

Mentor, Project SHORT

2020-2023

Speaker, Gateway High School and Fremont High School, High School Career Day

2020-2023

Mentor, Research Science Institute, MIT

2016-2019

Volunteer computer science instructor, Will Rogers High School, Tulsa, OK

2014-2015

SCIENTIFIC MENTORSHIP

New York University Undergraduate Students	2025-present
Tri-Institutional MD/PhD Students	2024-present
Weill Cornell Program in Biophysics, and Structural Biology PhD Students	2023-present
Weill Cornell Computational Biology and Medicine PhD Students	2023-present
Gerstner Sloan Kettering PhD Students	2023-present
Stanford Institutes of Medicine Summer Research Program	2022-2023
Stanford undergrad. and graduate students in: Genetics, Computer Science, and Immunology	2020-2023
MIT undergraduate research	2016-2020

PROFESSIONAL MEMBERSHIP

Member, American Association for Cancer Research (AACR)

Member, American Society of Human Genetics (ASHG)

Developer, Bioconductor Project

Supporting Member, R and Python Software Foundations

GUEST EDITOR

Genome Biology, Frontiers in Immunology

AD HOC PEER REVIEW - GRANTS

NIH Human Virome Study Section (x2), NCI IMAT Study Section, Dutch Research Council, Italian Science Fund, Austrian Science Fund, National Science Foundation, Simons Research Foundation, European Research Council (ERC), UK Medical Research Council, Swiss National Science Foundation, Wellcome Trust, Curebound Grants, Michelson Medical Research Foundation

AD HOC PEER REVIEW - JOURNALS

Nature, Cell, Science, PNAS, Nature Biotechnology, Nature Genetics, Nature Methods, Nature Medicine, Cell Systems, Nature Protocols, Science Advances, Nature Structural and Molecular Biology, Genome Research, BMC Bioinformatics, Nature Communications, Nucleic Acids Research, BMC Biology, Cancers, Communications Biology, The R Journal, Life Science Alliance, BioData Mining, BMC Medical Genomics, BMJ Open, Molecular Cancer, Frontiers in Medicine, Frontiers in Cell and Development Biology, Frontiers in Immunology, Cell Reports Medicine, Oxford Bioinformatics Molecular Systems Biology

INVITED TALKS AND PRESENTATIONS (2023-Present; @MSKCC)

2nd Annual EMBO Lineage Tracing and Dynamics Meeting , Sant Feliu de Guixols, Spain	September 2027
Arc Institute Invited Seminar Series , Palo Alto, CA	April 2027
9th Annual Oz Single Cell Conference , Victoria, Australia	March 2027
University of Washington Genome Sciences Seminar Series , Seattle, WA	March 2027
Baylor Department of Molecular and Cellular Biology Seminar Series , Houston, TX	February 2027
NYU Langone Cancer Center Seminar Series , New York, NY	November 2026
Vanderbilt Medical Center Genomics and Personalized Medicine Seminar Series , Nashville, TN	October 2026
25th Annual InterLymph Consortium Conference , Charleston, SC	October 2026
Stanford University Immunology Seminar Series , Stanford, CA	October 2026
International Mammalian Synthetic Biology Workshop, AiChE , Atlanta, GA	August 2026
Massachusetts General Hospital, Cancer Center Seminar Series , Boston, MA	June 2026
Boston Festival of Genomics Annual Meeting , Boston, MA	June 2026
American Society of Microbiology General Meeting , Washington, DC	June 2026

AstraZeneca Centre for Genomics Research Seminar Series , London, UK	April 2026
American Association for Cancer Research General Meeting , San Diego, CA	April 2026
3rd Annual Probing Human Disease Using Single-Cell Technologies , Cancun, Mexico	March 2026
54th Meeting of the Japanese Society of Immunology , Himeji, Japan	December 2025
University of Helsinki, Clinical and Applied Virology Course , Helsinki, Finland	October 2025
Mt. Sinai Department of Microbiology Seminar Series , New York, NY	October 2025
New York University Department of Biology Seminar Series , New York, NY	October 2025
FASEB: Mechanisms of mitochondrial DNA mutation and repair , Nashville, TN	June 2025
UCSF Cancer Biology Seminar Series , San Francisco, CA	April 2025
Osaka University Immunology Seminar Series , Osaka, Japan	March 2025
12th International Conference on HHV-6 and 7 , Himeji, Japan	March 2025
23rd Annual Workshop at Bellairs– Metagenomics Barbados	February 2025
University of Toronto, Medical Genetics Seminar Series , Houston, TX	November 2024
Baylor University, Genetics and Genomics Seminar Series , Toronto, ON	November 2024
Single-cell Genomics Conference 2024 , Corinthia, Greece	September 2024
Columbia University Intensive Workshop: Methods in Single-Cell Data , New York, NY	July 2024
Gordon Research Conference- Single-cell Cancer Biology , Hooksett, NH	June 2024
Innovations in Single Cell Omics , Barcelona, Spain	May 2024
PEGS - Antibody and Protein Engineering , Boston, MA	May 2024
Element Biosciences User Group Meeting , New York, NY	April 2024
New York Genome Center- Single-cell Genomics Day , New York, NY	March 2024
Single-cell Netherlands Seminar Series , Amsterdam, NL	March 2024
Immuno-Oncology 360 Summit , Brooklyn, NY	February 2024
2nd Annual Probing Human Disease Using Single-Cell Technologies , Cancun, Mexico	February 2024
Google DeepMind Invited Seminar Series , London, UK	November 2023
38th Annual Meeting, Society for Immunotherapy of Cancer , Chicago, IL	September 2023

PATENTS

GRANTED

1. **PCT/US2019/036583**; Lineage tracing using mitochondrial genome mutations and single-cell genomics
2. **PCT/US2021/035951**; Methods of detecting mitochondrial disease
3. **PCT/US2021/032702**; Lineage inference from single-cell transcriptomics
4. **PCT/US2023/025711**; Methods of Genetically Modifying Cells for Altered Codon-Anti-Codon Interactions
5. **PCT/US2023/030111**; Methods of assessing therapeutic T cells for latent and reactivated human herpesvirus 6
6. **PCT/US2023/025711**; Methods of genetically modifying cells for altered codon-anti-codon interactions
7. **PCT/US2025/020760**; Programmable Enrichment Via RNA Fish for Single-Cell RNA Analysis

PENDING

8. **PPA 63/992,879**; Single-cell sequencing for quantifying transfer RNAs
9. **PPA 63/910,627**; Sequence and structural determinants of efficacious de novo chimeric antigen receptors
10. **PPA 63/314,980**; Massively integrated analysis of single cell RNA sequencing data

Scientific Advisory Board

2025- Present

NeoSpliceTx (New York, NY)

Seed-stage therapeutics company

Scientific Co-Founder; Former Head of Platform Technology

2020- Present

Cartography Biosciences (South San Francisco, CA)

Clinical-Stage, Series B-backed cancer therapeutics company

BIBLIOGRAPHY

* Denotes equal contributions; ‡ Denotes corresponding author(s)

1. PUBLISHED PRIMARY RESEARCH ARTICLES (Peer-reviewed)

1A. Corresponding Author

1. S. Nyeo, E. Cumming, ..., S. Petrovski[‡], R. Dhindsa[‡], **C.A. Lareau[‡]**. (2026) Population-scale sequencing resolves determinants of persistent EBV DNA. *Nature*. DOI: 10.1038/s41586-025-10020-2.
2. A. Chow*, H. Chu*, ..., **C.A. Lareau[‡]**. (2026) Sequence and structural determinants of efficacious *de novo* chimaeric antigen receptors. *Nature Biomedical Engineering*. DOI: 10.1038/s41551-026-01790-9.
3. V. Liu, K. Sandor, ..., **C.A. Lareau[‡]**, A.T. Satpathy[‡]. (2026) Clonal lineage tracing of innate immune cells in human cancer. *Cancer Cell*. DOI: 10.1016/j.ccell.2026.05.006.
4. T. Abay, R. Stickels, ... , R. Chaligne[‡], A.T. Satpathy[‡], **C.A. Lareau[‡]**. (2025) Transcript-specific enrichment enables profiling of rare cell states via scRNA-seq. *Nature Genetics*. DOI: 10.1038/s41588-024-02036-7.
5. **C.A. Lareau[‡]**, *et al.* (2025) Cell type-specific purifying selection of synonymous mitochondrial DNA variation. *Proceedings of the National Academy of Sciences*. DOI: 10.1101/2023.04.23.537997v1.
6. **C.A. Lareau[‡]**, *et al.* (2023) Latent human herpesvirus 6 is reactivated in CAR T cells. *Nature*. DOI: 10.1038/s41586-023-06704-2.
7. **C.A. Lareau[‡]**, *et al.* (2023) Single-cell multi-omics reveals dynamics of purifying selection of pathogenic mitochondrial DNA across human immune cells. *Nature Genetics*. DOI: 10.1038/s41588-023-01433-8.
8. **C.A. Lareau[‡]**, *et al.* (2023) Mitochondrial single cell ATAC-seq for high-throughput multi-omic detection of mitochondrial genotypes and chromatin accessibility. *Nature Protocols*. DOI: 10.1038/s41596-022-00795-3.
9. **C.A. Lareau^{*,‡}**, L.S. Ludwig^{*,‡}, C. Muus, *et al.* (2021) Massively parallel single-cell mitochondrial DNA genotyping and chromatin profiling. *Nature Biotechnology*. DOI: 10.1038/s41587-020-0645-6.
10. **C.A. Lareau[‡]**, S. Ma, F.M. Duarte, J.D. Buenrostro[‡]. (2020) Inference and effects of barcode multiplets in droplet-based single-cell assays. *Nature Communications*. DOI: 10.1038/s41467-020-14667-5.

1B. (Co-) first author

11. L.S. Ludwig*, **C.A. Lareau***, *et al.* (2022) A Congenital Anemia Dissociates the Pleiotropic Functions of Master Transcription Factor GATA1. *Blood*. DOI: 10.1182/blood.2021013753.
12. E. Mimitou*, **C.A. Lareau***, K.Y. Chen* *et al.* (2021) Scalable, multimodal profiling of chromatin accessibility, RNA, and protein levels in single cells. *Nature Biotechnology*. DOI: 10.1038/s41587-021-00927-2.

13. J. Wang*, **C.A. Lareau***, *et al.* (2021) Single-Cell Multiomics Defines Tolerogenic Extrathymic Aire-Expressing Populations with Unique Homology to Thymic Epithelium. *Science Immunology*. DOI: 10.1126/sciimmunol.abl5053.
14. N. Schmidt*, **C.A. Lareau***, H. Keshishian* *et al.* (2021) A direct RNA-protein interaction atlas of the SARS-CoV-2 RNA in infected human cells. *Nature Microbiology*. DOI: 10.1101/2020.07.15.204404.
15. M. A. Walker*, **C.A. Lareau***, L.S. Ludwig*, *et al.* (2020) Purifying Selection against Pathogenic Mitochondrial DNA in Human T cells. *New England Journal of Medicine*. DOI: 10.1056/NEJMoa2001265.
16. D. Vuckovic*, E. Bao*, P. Akbari*, **C.A. Lareau***, *et al.* (2020) The Polygenic and Monogenic Basis of Blood Traits and Diseases. *Cell*. DOI: 10.1101/2020.02.02.20020065.
17. L.S. Ludwig*, **C.A. Lareau***, J.C. Ulirsch*, *et al.* (2019) Lineage tracing in humans enabled by mitochondrial mutations and single cell genomics. *Cell*. DOI: 10.1016/j.cell.2019.01.022.
18. **C.A. Lareau***, F.M. Duarte*, J.G. Chew*, *et al.* (2019) Droplet-based combinatorial indexing for massive-scale single-cell chromatin accessibility. *Nature Biotechnology*. DOI: 10.1038/s41587-019-0147-6.
19. **C.A. Lareau**, L.S. Ludwig, V.G. Sankaran. (2019) Longitudinal assessment of clonal mosaicism in human hematopoiesis via mitochondrial mutation tracking. *Blood Advances*. DOI: 10.1182/bloodadvances.2019001196.
20. J.C. Ulirsch*, **C.A. Lareau***, E.L. Bao*, *et al.* (2019) Interrogation of human hematopoiesis at single-cell and single-variant resolution. *Nature Genetics*. DOI: 10.1038/s41588-019-0362-6.
21. E.L. Bao*, **C.A. Lareau***, C. Brugnara, I.R. Fulcher, *et al.* (2019) Heritability of fetal hemoglobin, white cell count, and other clinical traits from a sickle cell disease family cohort. *American Journal of Hematology*. DOI: 10.1002/ajh.25421.
22. L.S. Ludwig*, **C.A. Lareau***, *et al.* (2019) Transcriptional States and Chromatin Accessibility Underlying Human Erythropoiesis. *Cell Reports*. DOI: 10.1016/j.celrep.2019.05.046.
23. **C.A. Lareau** and M.J. Aryee. (2018) hichipper: A preprocessing pipeline for assessing library quality and DNA loops from HiChIP data. *Nature Methods*. DOI: 10.1038/nmeth.4583.
24. **C.A. Lareau**, C.F. DeWeese *et al.* (2017) Polygenic risk assessment reveals pleiotropy between sarcoidosis and inflammatory disorders in the context of genetic ancestry. *Genes and Immunity*. DOI: 10.1038/gene.2017.3
25. **C.A. Lareau** and M.J. Aryee. (2017) diffloop: a novel computational framework for identifying and functionalizing differential topological features in DNA. *Bioinformatics*. DOI: 10.1093/bioinformatics/btx623.
26. **C.A. Lareau**, B.C. White, *et al.* (2016) An interaction quantitative trait loci tool implicates epistatic functional variants in an apoptosis pathway in smallpox vaccine eQTL data. *Genes and Immunity*. DOI: 10.1038/gene.2016.15.
27. **C.A. Lareau**, B.C. White, C.G. Montgomery, and B.A. McKinney. (2015) Common Variants Modulate Differential Correlation Structures in Gene Expression Data. *Frontiers in Genetics*. DOI: 10.3389/fgene.2015.00312.
28. **C.A. Lareau**, I.A. Adrianto, *et al.* (2015) Fine mapping of Chromosome 15q25 in Sarcoidosis Implicates Zinc Finger Protein 592 in Patients with Neurological Involvement. *Annals of Translational and Clinical Neurology*. DOI: 10.1002/acn3.229.
29. **C.A. Lareau**, B.C. White, A.L. Oberg, and B.A. McKinney. (2015) Differential co-expression network centrality and machine learning feature selection for identifying susceptibility hubs in networks with scale-free structure. *BioData Mining*, DOI: 10.1186/s13040-015-0040-x.
30. N.A. Davis*, **C.A. Lareau***, *et al.* (2013) Encore: Genetic association interaction network centrality pipeline and application to SLE exome data. *Genetic Epidemiology*. DOI: 10.1002/gepi.21739.

1C. Co-authored

31. B. Daniel, ..., **C.A. Lareau**, K. Yost, H. Chang, A. Satpathy. (2024) Regulation of immune signal integration and memory by inflammation-induced chromosome conformation. *PNAS*. DOI: 10.1073/pnas.2600693123.
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35. K. Yu, ..., **C.A. Lareau**, N. Agar, T. Merghoub, V. Tabar. (2026) A pathogenic subpopulation of human glioma associated macrophages linked to glioma progression *Cancer Cell*. DOI: 10.1101/2025.02.12.637857.
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37. L. Siraj, R. Castro, H. Dewey, S. Kales, T. Nguyen, M. Kanai, D. Berenzy, K. Mouri, Q. Wang, Z. McCaw, S. Gosai, F. Aguet, R. Cui, C. Vockley, **C.A. Lareau**, *et al.* (2026) Functional dissection of complex and molecular trait variants at single nucleotide resolution. *Nature*. DOI: 10.1038/s41586-026-10121-6
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44. D. Luan, S. DeWolf, T. Fei, S. Raj, G. Shah, **C.A. Lareau**, *et al.* (2024) Dynamics of immune reconstitution and impact on outcomes of CAR-T Cell products in large B-cell lymphoma. *Blood Cancer Discovery*. DOI: 10.1158/2643-3230.BCD-24-0163.
45. A. Obers, ..., **C.A. Lareau**, *et al.* (2024) Retinoic acid and TGF- β orchestrate organ-specific programs of tissue residency; *Immunity*. DOI: 10.1016/j.immuni.2024.09.015.
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47. F.A. Buquicchio, ... **C.A. Lareau**, A.T. Satpathy, L. Mackay. (2024) Distinct epigenomic landscapes underlie tissue-specific memory T cell differentiation. *Immunity*. DOI: 10.1016/j.immuni.2024.06.014

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